Experiment 3: Visual-Inertial Odometry With Dynamic-Environment Filtering

Raw write-up for Chapter 4, Section 4.3

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Purpose of this document. This is the raw, complete write-up of Experiment 3, written before the thesis chapter; Section 4.3 of the final report is extracted from it. It holds both result slots, every table and figure, and the analysis. Every number traces to a file under `output/` named in the "Reporting rules" paragraph of its slot. Open concerns are set in red where they arise. The extreme-world campaign of September and October 2026 (scripted kinematic props, 66 repeats and 14 filtered runs) was retired on 6 November 2026 with its datasets and is not reported; its artefacts are under `trash/output/`.

4.3.1 Simulation

Status: partial. Unfiltered baselines only. Twenty runs on the four `dataset_allsensor_realistic` recordings (two layouts, each without and with physically simulated traffic): VINS-Fusion once per recording, ORB-SLAM3 with a CPU or GPU feature extractor and its loop closer active or disabled. Traffic is noted as a condition, not tested as a factor. No filtered run exists in the current dataset set, so the filtering hypothesis is untested, and the latency and resource measurement is taken from eighteen repeated runs on the two mapping recordings: closure-free ORB-SLAM3 with each front end, and VINS-Fusion, three repeats of every configuration.

Objective. Determine whether masking dynamic-object features improves localisation, at what compute cost, and whether tracking survives it. In the present state of the slot the question answered is the precondition: how each unfiltered pipeline behaves on the realistic warehouse, with and without traffic, and which of its own switches (front end, loop closer) decides its error.

Process. Every run replayed at rate 1.0 from the bag with the node live on the laptop, the deployment condition; no detector ran. VINS-Fusion ran once per recording (run `cpu`). ORB-SLAM3 ran four times per recording: the CPU or the GPU feature extractor (fork patches 10/10b, runs `*_cpu` and `*_gpu`), each with the loop closer active (`loopclosing_*`) or disabled (`noloopclosing_*`), all on 6 November 2026 with the same node binary and configuration (`config/wil_sim/stereo_imu.yaml`). One run per cell, so no within-condition spread exists.

Datasets. `realistic_000` (U layout) and `_001` (I layout) are the static twins; `_002` (U) and `_003` (I) carry ten and twelve moving props, pallet jacks on pick–carry–drop jobs and walking people. Routes are 103 to 171 m over 80 to 126 s (the dataset table of the Experiment 1 write-up); no two recordings share a route.

Reporting rules. Every run is reported on its own. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed, in three dimensions; RPE at 1 s is translational over pairs that span no jump (a step above $10\,\mathrm{m\,s^{-1}}$ between consecutive poses). The loop-closer state is read from `loop_closing.csv` (sub-millisecond rows mean a disabled closer), not from the run name, and every closure the closer accepted is listed with its time and correction. Ground truth is the bag's `/ground_truth/odometry`, moved from the model origin to the IMU point both estimators report (lever arm 0.338 m forward, 0.192 m up) before alignment; the figures draw that moved path. "Kept" is the fraction of the bag's stereo pairs the node processed, so frames lost before the node's own counter count as lost. The onset analysis uses the start-anchored planar error against a threshold of three times the estimator's median error over its two static runs (VINS-Fusion 1.07 m, ORB-SLAM3 1.31 m). A filtered run, when one exists, is paired with the baseline of the same estimator, front end and loop-closer state, and the pair is admissible only under three gates: an active filter (`detection_match_rate` ≥ 0.98), processed frames and duration within 5 % of the baseline, and both runs on the same node binary. Sources: `output/compare/simulation/exp-realistic-ab/` (`runs_metrics.csv`, `baseline_table.tex`), `output/compare/simulation/exp3-timelines/` (`timelines.csv`, `thresholds.csv`), run directories under `output/output_orb/simulation/` and `output/output_vins/simulation/`. The run that first occupied `realistic_002/loopclosing_gpu` had been launched on the `realistic_003` bag with the wrong output path (its manifest names that bag and world, and it received 3 535 stereo pairs where `realistic_002` holds 2 409); it is set aside under `trash/output/misfiled_20261106/` and the cell was re-run on the right bag on the evening of 6 November.

Results. VINS-Fusion holds 0.22–0.66 m aligned ATE on the four recordings with no jump and 1.2–3.6° of rotation error; its worst recording is the U layout with traffic, 0.66 m against 0.29 m on that layout's static twin, while on the I layout traffic changes nothing (0.27 against 0.22 m). ORB-SLAM3 spans 0.14–6.02 m: eleven of its sixteen runs lie between 0.14 and 0.52 m and five diverge, and which five is decided by the loop closer and the recording, not by the front end (table 3.1, figures 3.1 and 3.2). Neither front end loses tracking on any recording, with or without traffic: zero tracking-loss events and zero resets in twenty runs.

Table 3.1: Unfiltered arms on the four realistic-warehouse recordings, 6 November 2026, replay rate 1.0. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed; RPE is translational, at $\Delta = 1\,\mathrm{s}$ over pairs that do not span a jump; "Jumps" counts steps above $10\,\mathrm{m\,s^{-1}}$; "Kept" is the fraction of the bag's stereo pairs the estimator processed (frames lost before the node's own queue count as lost). Loop-closer state is read from `loop_closing.csv`, not from the run name. Static and dynamic rows of one layout share a world geometry and differ in traffic only.

Recording	Layout	Scene	Estimator	ATE [m]	Rot. [°]	$\mathrm{RPE}_{1\,\mathrm{s}}$ [m]	Jumps	Kept	Resets
`realistic_000`	U	static	ORB-SLAM3	0.338	13.74	0.083	2	0.72	0
			ORB-SLAM3 GPU	0.293	12.80	0.079	3	0.99	0
			ORB-SLAM3 (no LC)	0.483	14.05	0.090	2	0.74	0
			ORB-SLAM3 GPU (no LC)	0.523	12.84	0.078	3	0.99	0
			VINS-Fusion	0.290	1.85	0.084	0	0.77	0
`realistic_001`	I	static	ORB-SLAM3	6.021	16.92	0.094	6	0.73	0
			ORB-SLAM3 GPU	4.780	18.98	0.092	7	0.99	0
			ORB-SLAM3 (no LC)	0.496	16.05	0.093	1	0.74	0
			ORB-SLAM3 GPU (no LC)	0.400	15.13	0.088	2	0.97	0
			VINS-Fusion	0.221	1.18	0.089	0	0.78	0
`realistic_002`	U	dynamic	ORB-SLAM3	1.393	16.99	0.078	4	0.77	0
			ORB-SLAM3 GPU	0.400	16.04	0.078	4	0.98	0
			ORB-SLAM3 (no LC)	0.293	17.12	0.080	2	0.83	0
			ORB-SLAM3 GPU (no LC)	0.170	16.08	0.083	1	0.99	0
			VINS-Fusion	0.663	3.57	0.116	0	0.88	0
`realistic_003`	I	dynamic	ORB-SLAM3	0.139	13.52	0.075	2	0.94	0
			ORB-SLAM3 GPU	0.504	13.16	0.076	3	0.99	0
			ORB-SLAM3 (no LC)	1.755	14.22	0.080	2	0.88	0
			ORB-SLAM3 GPU (no LC)	1.911	13.53	0.083	3	0.99	0
			VINS-Fusion	0.268	1.51	0.071	0	0.87	0

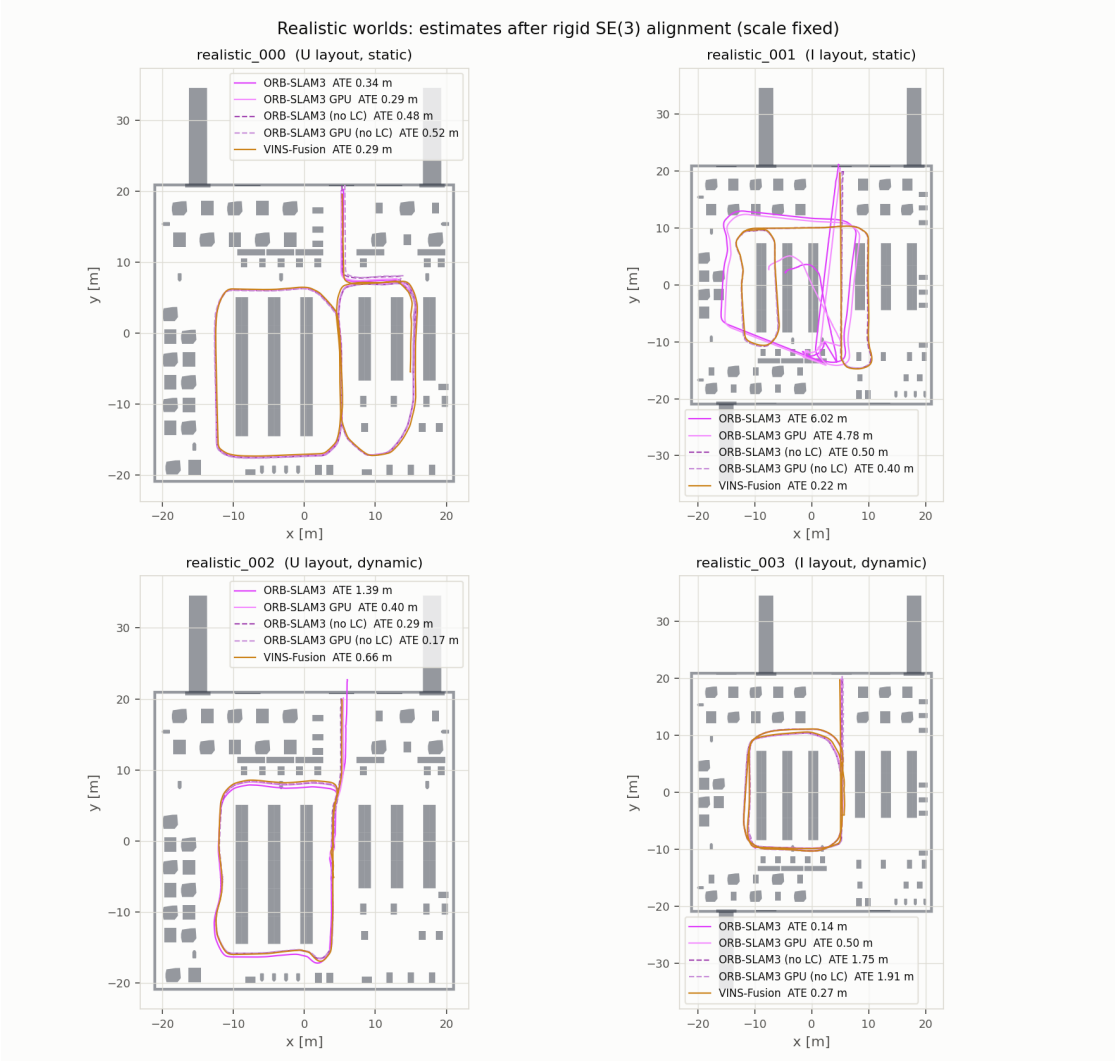


Figure 3.1: Unfiltered estimates on the four realistic-warehouse recordings after rigid $SE(3)$ alignment with scale fixed, on the warehouse footprint. Top: the static twins; bottom: the same layouts with traffic. Solid magenta is ORB-SLAM3 with its loop closer active, dashed with it disabled, lighter for the GPU front end; orange is VINS-Fusion. The two excursions on `realistic_001` are the loop-closing runs after their false closure; the 1.4 m run on `realistic_002` is the CPU loop-closing run; the closure-free runs on `realistic_003` lie on the route in plan and diverge vertically.

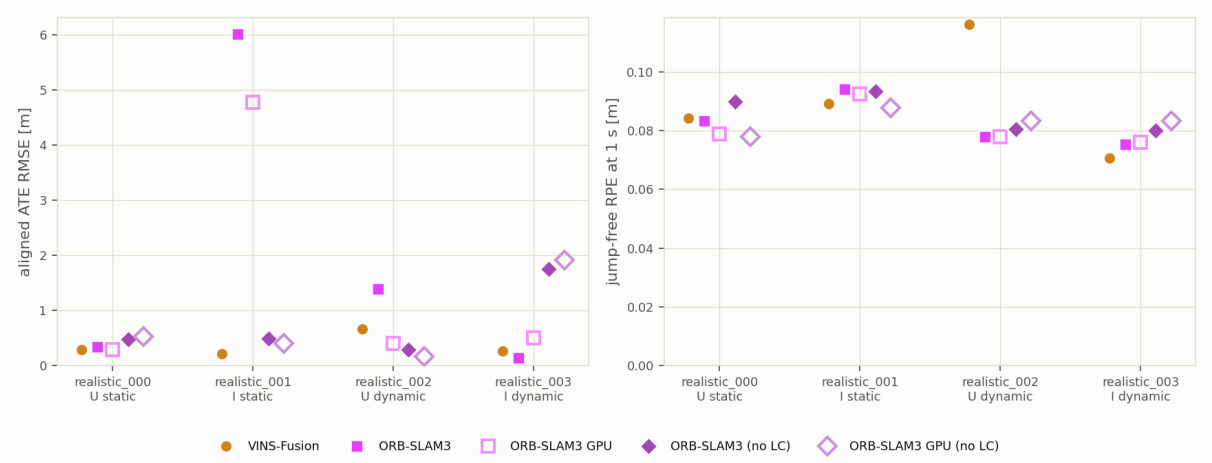


Figure 3.2: Aligned ATE (left) and jump-free translational RPE at 1 s (right) of every unfiltered run, one marker per run; filled markers are the CPU front end, open markers the GPU front end of the same configuration. The relative error is 0.07–0.12 m for every run, so the ATE differences are discrete events, not tracking quality.

Analysis: the loop closer decides, in both directions. On the static I layout both loop-closing runs accept one closure at 73.2 s, in the heading change at 76 % of the route, with a correction of 14.7 m (CPU) and 13.3 m (GPU) and 1.9–2.8°; a global bundle adjustment follows and the trajectory is 6.02 and 4.78 m off for the rest of the run, where the two closure-free runs on the same recording hold 0.50 and 0.40 m (figure 3.4). A route with half a metre of accumulated error cannot need a fourteen-metre correction, so the match is false: the I layout's long parallel aisles present the same rack faces twice, and the matched keyframes (97 against 255, 104 against 272) are one aisle apart. The same candidate was then re-proposed four times and rejected by the inertial geometry check, after the damage. That both front ends accept it at the same frame with the same correction says the place-recognition stage, not the extractor, is where the fault lies.

On the I layout with traffic the sign reverses. Both loop-closing runs accept one closure at 56.5 s (48 % of the route) with a 0.33 m correction and hold 0.14 and 0.50 m; the closure-free runs read 1.76 and 1.91 m, yet their planar error never exceeds 0.7 m (figure 3.6). The difference is vertical: the closure-free estimates climb to 4.7 and 5.1 m above the floor by the end of the run while the loop-closing runs stay within 0.6 and 1.4 m, and the three-dimensional ATE reports that climb. The GPU loop-closing run on this recording also rejected 148 further candidates by inertial geometry, every one of them after its single accepted closure. On the U layout the closer is nearly inert: two accepted closures per run on `realistic_000` with corrections of 0.07–0.12 m, one of 0.26–0.33 m per run on `realistic_002`, and the eight U-layout ORB-SLAM3 runs within 0.17–0.52 m except the CPU loop-closing run on `realistic_002`, whose 1.39 m is an offset present from 6.5 s on, before any closure (figure 3.5); with four jumps and the lowest relative error on the recording it is an initialisation event, not a tracking one, and the GPU run of the same configuration, which accepts the same closure at 65 s, does not reproduce it (0.40 m).

Analysis: the front end changes throughput, not accuracy. The GPU extractor processes 97–99 % of each bag's frames where the CPU extractor processes 72–94 %, and its tracking median is 30–37 ms against 49–55 ms (from the per-run `performance.csv`). Across the eight CPU/GPU pairs with the same closer state the ATE differs by at most 0.16 m in five; the three larger differences (−1.2, −1.0 and 0.4 m) are each one run's discrete event, in both directions, and the two divergence patterns above occur in both front ends alike. At this frame budget the extractor is not the limiting stage of accuracy on these recordings.

Analysis: traffic as a condition. VINS-Fusion is 0.37 m worse on the U layout with traffic than on its static twin and 0.05 m worse on the I layout; its relative error on `realistic_002` is the highest of any

run (0.116 m) and its error trace carries two waves of 0.7 m at 25 and 45 s on the western and southern aisles, where the pallet jacks work, that no static run shows (figure 3.5). ORB-SLAM3's closure-free runs are *better* on the U traffic recording (0.29, 0.17 m) than on its static twin (0.48, 0.52 m). With one run per cell and routes that differ between twin and traffic recording, neither number is a traffic effect; what the slot establishes is that physically simulated traffic degrades the tracker that follows features longest, and breaks neither.

Analysis: the rotation column. ORB-SLAM3's 13–19° rotational RMSE is one event on every run: the first jump lies at 2.5–4.8 s (table 3.2), where inertial initialisation re-aligns the world frame to gravity, and after it the clean rotational RPE is 0.07–0.33°. The jump detector is translational, so the column includes the event; the median is the number to read.

Table 3.2: Realistic world, unfiltered runs: onset of sustained error and its place on the route, per run. Onset is the first time the start-anchored position error stays above 3× the estimator's median error over its static runs (VINS-Fusion 1.07 m, ORB-SLAM3 1.31 m) for the rest of the run; "where" is the route segment and the share of the route completed at that time; the first jump is a step above 10 m s^{-1} between consecutive poses. "never" means the error stayed below the threshold for the whole run.

Recording	Estimator	LC	Onset [s] (where)	First jump [s]
`realistic_000`	ORB-SLAM3	active	never	4
	ORB-SLAM3	disabled	never	4
	ORB-SLAM3 GPU	active	never	4
	ORB-SLAM3 GPU	disabled	60 (straight 48 %)	4
	VINS-Fusion	—	never	—
`realistic_001`	ORB-SLAM3	active	73 (turn 78 %)	5
	ORB-SLAM3	disabled	never	5
	ORB-SLAM3 GPU	active	73 (turn 78 %)	5
	ORB-SLAM3 GPU	disabled	never	5
	VINS-Fusion	—	never	—
`realistic_002`	ORB-SLAM3	active	6 (straight 6 %)	3
	ORB-SLAM3	disabled	never	3
	ORB-SLAM3 GPU	active	never	4
	ORB-SLAM3 GPU	disabled	never	3
	VINS-Fusion	—	never	—
`realistic_003`	ORB-SLAM3	active	never	3
	ORB-SLAM3	disabled	never	3
	ORB-SLAM3 GPU	active	never	3
	ORB-SLAM3 GPU	disabled	never	3
	VINS-Fusion	—	never	—

Filtered arms. untested. The current dataset set holds no filtered run: the October 2026 filtered runs on these recordings were retired on 6 November 2026 with the rest of the earlier campaign, and the filter has not been re-run under the present pipeline (GPU front end, patch-15 initialisation switches). When it is, each filtered run pairs with the baseline of the same estimator, front end and closer state under the three gates above, and the loop-closer finding here fixes the pairing that matters: on the I layout the comparison must be made with the closer disabled, or the false closure will dominate both arms.

Latency and resources. By the design decision of 6 November 2026 the real-time and resource measurement of this experiment is taken only on the two mapping recordings, each configuration run

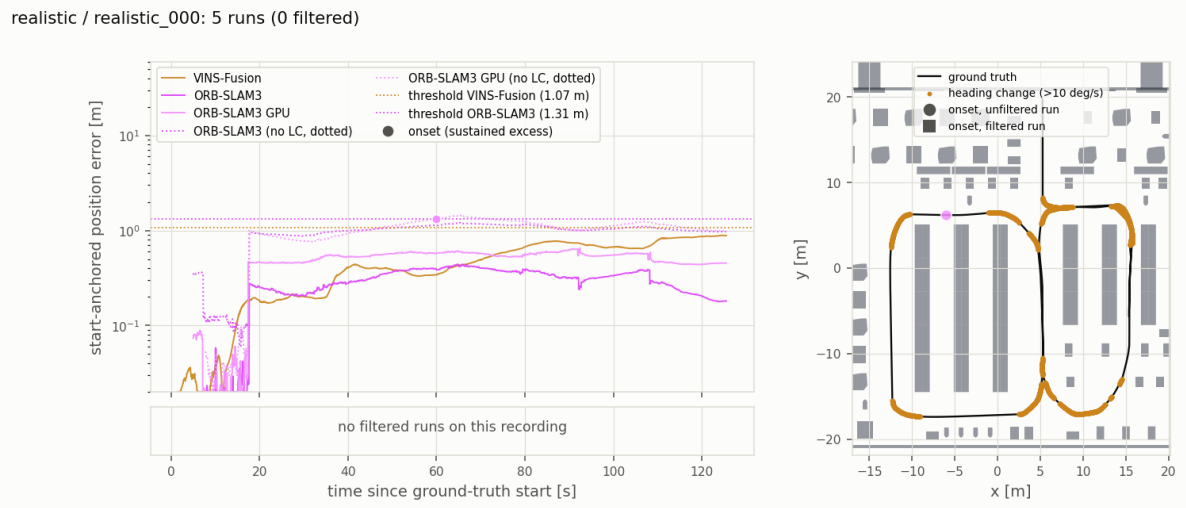


Figure 3.3: `realistic_000` (U, static), every run. Left: start-anchored planar error on a log axis, solid for loop closing active, dotted for disabled, lighter for the GPU front end, with each estimator's threshold; the dot marks the onset of sustained excess. Right: the route with heading changes in orange and every onset placed on it. The one onset is the GPU closure-free run at 60 s, a sustained 1.3 m offset on the straight, not a jump.

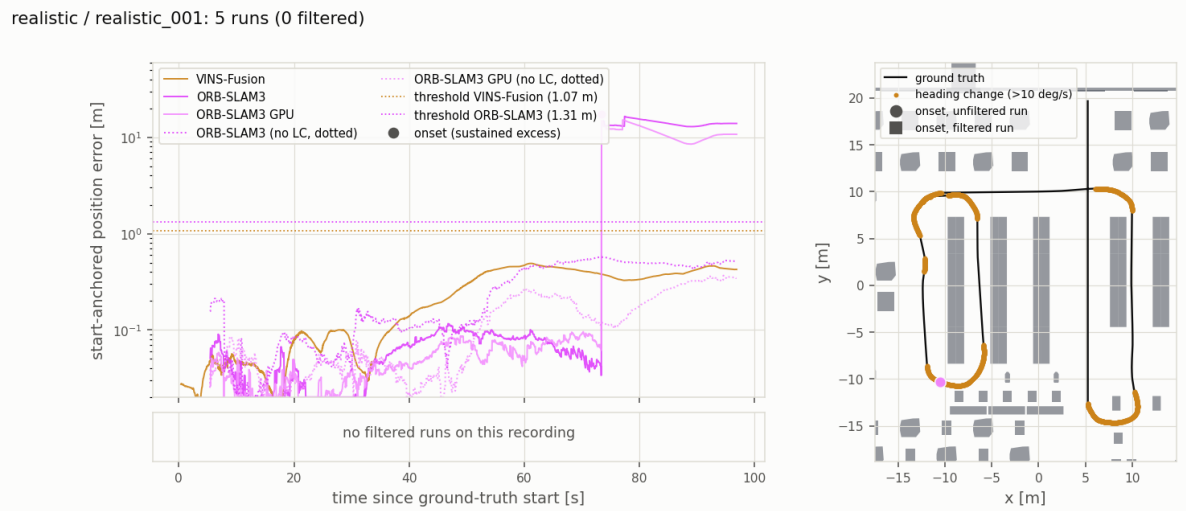


Figure 3.4: `realistic_001` (I, static), every run; panels as in figure 3.3. Both loop-closing runs step to 14 m of error at 73 s, in the heading change at 76 % of the route, at the single accepted closure; the closure-free runs and VINS-Fusion never cross.



Figure 3.5: `realistic_002` (U, traffic), every run; panels as in figure 3.3. The CPU loop-closing run carries a 1.5 m offset from 6.5 s, which the GPU loop-closing run does not share; VINS-Fusion's two 0.7 m waves at 25 and 45 s lie on the aisles where the pallet jacks work.

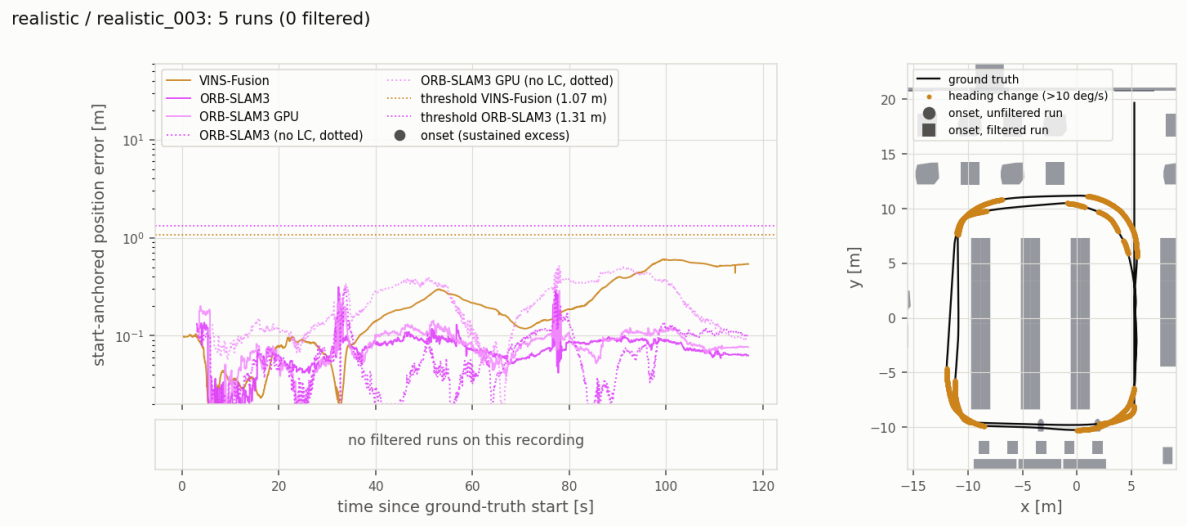


Figure 3.6: `realistic_003` (I, traffic), every run; panels as in figure 3.3. No run crosses its threshold in the plane; the 1.8 and 1.9 m ATE of the closure-free runs is vertical drift, which the single accepted closure of the loop-closing runs removes.

three times: `dataset_allsensor_realistic_map` (U layout, 286 s, 8 725 stereo pairs) and `dataset_allsensor_realistic_01_map` (I layout, 315 s, 9 552 pairs). Eighteen runs exist, six configurations of three repeats each at replay rate 1.0 with no detector: ORB-SLAM3 with the CPU and with the GPU front end, loop closer disabled, and VINS-Fusion (table 3.3). The repeats are tight, front-end medians within 1–7 ms of their mean and peak memory within 30 Mi B.

The two pipelines divide the cost differently, and neither is cheaper than the other outright. VINS-Fusion optimises in 44 ms where ORB-SLAM3's local mapping takes 211–406 ms, and holds its state in a quarter of the memory, but spends more than twice the processor time: about eight cores of one against three and a half. Their front ends are within 6 ms of each other, and every configuration keeps 75–96 % of the bag's frames, so the choice between them is a memory-against-cores trade, not a speed one.

Between ORB-SLAM3's two front ends the recordings disagree. On the U recording the GPU extractor behaves as the single runs showed, tracking in 37 against 42 ms and keeping 96 against 93 % of frames; on the I recording the sign reverses, 52 against 44 ms and 75 against 91 % kept, with the local-mapping median at 406 ms. VINS-Fusion reads the same on both recordings (front end 38.1 and 38.6, back end 44.3 and 44.5 ms), so the recordings themselves do not impose a difference of that size.

[Confound: run order] The six GPU runs were made consecutively from 21:04 to 22:04 and the six CPU runs from 23:27 to 00:18, so the I-recording GPU repeats were the fourth to sixth GPU runs of a session on a laptop that throttles at 90 °C; the front-end record notes that timing comparisons need a cool-start, interleaved order. The six VINS-Fusion runs are a separate session of 7 November and are not a control for that state. The U-recording figures are the ones to quote until the I recording is repeated interleaved.

The per-run resources of the twenty accuracy runs above are generated beside this table (`realistic_resources` and are not reported, except the frame retention already in table 3.1.

Table 3.3: Real-time and resource measurements on the two mapping recordings, replay rate 1.0, no detector: mean ± SD over n repeats of each configuration. Front-end and back-end are each run's median module latency (ORB-SLAM3: tracking and local mapping; VINS-Fusion: feature tracking and back-end optimisation); throughput is processed frames per second of sensor time; "Kept" is the processed fraction of the bag's stereo pairs; CPU is process time over wall time, in percent of one core; RSS the peak resident set. The loop-closer state is read from `loop_closing.csv`.

Recording	Estimator	LC	n	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept [%]	CPU [%]	Peak RSS [Mi B]
`realistic_map`	ORB-SLAM3	disabled	3	42.0 ± 3.3	211.3 ± 14.8	28.3 ± 0.9	93 ± 3	379 ± 11	1732 ± 15
	ORB-SLAM3 GPU	disabled	3	36.7 ± 1.2	284.2 ± 14.7	29.1 ± 0.4	96 ± 1	332 ± 7	1794 ± 19
	VINS-Fusion	—	3	38.1 ± 3.3	44.3 ± 5.0	27.5 ± 2.7	91 ± 9	819 ± 10	479 ± 15
`realistic_01_map`	ORB-SLAM3	disabled	3	43.9 ± 1.0	220.4 ± 12.6	27.5 ± 0.3	91 ± 1	381 ± 6	1816 ± 29
	ORB-SLAM3 GPU	disabled	3	51.9 ± 3.8	405.7 ± 22.6	22.9 ± 1.4	75 ± 4	361 ± 9	1719 ± 21
	VINS-Fusion	—	3	38.6 ± 0.8	44.5 ± 1.5	27.4 ± 0.5	90 ± 2	814 ± 5	497 ± 11

Limitations, each stated once. *n:* one run per cell; the only run-to-run figure is the agreement of the two front ends on where and how the closer fails. *Route and traffic confounded:* no two recordings share a route. *Three-dimensional ATE:* the metric charges vertical drift to a floor robot; the planar onset table is the complement, not a replacement. *Deployment rate on one laptop:* frame retention and every latency figure are properties of this machine at rate 1.0, and the mapping-bag repeats were not interleaved between front ends.

[Open: the false closure is diagnosed from its correction, not from the images] The 14 m correction and the one-aisle keyframe separation identify the `realistic_001` closure as false, and both front ends agree; the matched image pair has not been inspected, and the two April-style guards that would catch it (a covisibility-consistency window, or a maximum correction against the inertial drift bound) are untested. Until one is, the finding is a diagnosis, not a fix.

Summary. On the realistic warehouse neither pipeline loses tracking, with or without traffic. VINS-Fusion holds 0.22–0.66 m with no jump and is worst on the U layout with traffic; ORB-SLAM3 holds 0.14–0.52 m in eleven runs of sixteen, and its five divergences are the loop closer's: a false closure on the I layout's repeated aisles that puts both front ends 4.8–6.0 m off, a vertical drift on the same layout with traffic that only the closer corrects, and one initialisation offset. The front end changes throughput (97–99 % of frames kept against 72–94 %) and not accuracy. The filtering hypothesis is untested in the current dataset set. On the mapping recordings VINS-Fusion optimises five to nine times faster than ORB-SLAM3's local mapping and in a quarter of the memory, for more than twice the processor time; within ORB-SLAM3 the GPU front end tracks in 37 ms and keeps 96 % of frames against the CPU front end's 42 ms and 93 % on the U layout, in three repeats each, and the I-layout repeats contradict that and were run in a thermally confounded order.

Future plan. Run both pipelines filtered on `realistic_002` and `_003` with the detector live, paired under the three gates and with the closer disabled on the I layout; add a correction bound to the loop closer and test it on `realistic_001`; repeat the mapping-recording runs interleaved between all three configurations from a cool start, so that the thermal state is shared rather than confounded with the configuration; and record a second route in each layout so that traffic and route are separable.

4.3.2 Real robot

Status: partial. Four real recordings with an independent planar reference. The first, `dataset_realv6_000` (31 October 2026), carries five unfiltered runs whose raw data exist: ORB-SLAM3 stereo-inertial with GPU or CPU front end and loop closing on or off, and VINS-Fusion once. ORB-SLAM3 tracks the whole route in all four configurations at a provisional 0.46 to 1.03 m planar ATE, the GPU front end at 0.46–0.52 m; VINS-Fusion resets 217 times on IMU-stream gaps and has no valid run. No filtered run exists, the `realv6` tracker export fails the pre-registered validity rules, the three `realv7` exports pass them and all twelve ORB-SLAM3 runs on them score after a frame repair in the scorer (0.25–0.27 m on `000`), but the IMU stayed uninitialised for most of nine runs (an upstream map reset at the initialisation, then a starved re-initialisation), VINS-Fusion has no valid run, and the marker-to-IMU lever arm is unmeasured, so the filter hypothesis remains untested on the real robot and no headline is claimed.

Objective. Establish whether the two unfiltered pipelines track the real robot against an independent reference, as the precondition for a filtered-versus-unfiltered test on real imagery.

Process. The bag was replayed into each node at real time on the afternoon of 3 November 2026; no detector ran. ORB-SLAM3 used `config/wil/stereo_imu.yaml` with the repaired library (fork patches 11–14b, node and library built alike), the 3 November stereo extrinsic and `IMU.NoiseGyro` 0.1 with `IMU.NoiseAcc` 1.0, ten times the values that had been copied from the VINS file, in four configurations: GPU or CPU feature extractor, loop closing on or off. VINS-Fusion used the same file with the gravity norm of an August recording. Earlier runs on this bag were overwritten by these and are not reported. The optical tracker export was scored with the real-robot procedure of Experiment 1.

Dataset. 220 s, 6 601 stereo pairs at 30 Hz, RealSense IMU at 200 Hz on the inherited topic name, wheel-speed feedback, a figure-of-eight of about 27 m at a median 0.15 m s^{-1} on the Gensurv floor, no scripted moving objects. Tracker export: 277 samples, median spacing 0.6 s, 66 gaps above 1 s, one sample 5.2 m off the path at 184.9 s.

Reporting rules. `script/eval_real_gt.py`: rigid SE(2) alignment with scale fixed, lever arm set to zero and unmeasured, every value provisional. The export fails the tool's gap and step rules, so the tool labels each run *failed (ground truth invalid)*; the values below are diagnostics. All are scored with the one off-path sample removed, an edit recorded here and applied nowhere else; table 3.5 shows what that sample does to one run. The bag was not on the machine, so the time offset is pinned to the value the wheel-speed cross-correlation found on 2 November and the manifests record the substitution. Sources: `output/compare/real/realv6_1103pm_*/`, `output/compare/real/realv6-insights/`, run directories under `output/output_orb/real/dataset_realv6_000/` and `output/output_vins/real/dataset_realv6_000/`.

Results. ORB-SLAM3 follows the whole 218 s route in every configuration: 0.52 and 0.46 m with the GPU extractor (loop closing on and off), 0.62 and 1.03 m with the CPU extractor, which drops two thirds of the frames at real time (table 3.4, figure 3.7). VINS-Fusion tracked 18 s before 217 IMU-gap resets took over (table 3.7) and has no entry in the accuracy table.

Table 3.4: ORB-SLAM3 on `dataset_realv6_000` against the cleaned tracker truth, 3 November 2026 afternoon series: repaired library, replay rate 1.0, 3 November stereo extrinsic, `IMU.NoiseGyro` 0.1 and `IMU.NoiseAcc` 1.0, GPU or CPU feature extractor, loop closing (LC) on or off. Aligned planar ATE after a rigid SE(2) fit with scale fixed; RPE at 1 s translational; path ratio = estimate / truth; resets = `local_mapping_bad_imu` map resets; dropped = camera frames the node's queue discarded of about 6 480 received. Lever arm tracker-marker to IMU unmeasured (zero applied); time offset pinned to −4.14 s; all values provisional under the scorer's validity rules.

Run	ATE [m]	median [m]	max [m]	$RPE_{1\,s}$ [m]	path ratio	resets	dropped
ORB-SLAM3, GPU, LC on	0.52	0.44	1.11	0.057	0.85	2	2420
ORB-SLAM3, CPU, LC on	0.62	0.48	1.38	0.110	0.88	2	4363
ORB-SLAM3, GPU, LC off	0.46	0.36	0.99	0.056	0.87	0	3201
ORB-SLAM3, CPU, LC off	1.03	0.89	1.83	0.223	0.96	3	4204

Table 3.5: The same ORB-SLAM3 run (GPU, LC on) scored against the tracker export as delivered and with one sample removed (the 5.2 m excursion at 184.9 s). The time offsets are the values the wheel-speed cross-correlation found on each export on 2 November (peak correlation 0.10 raw, 0.61 cleaned), applied here as pinned values. The cleaned truth still fails the gap rule (66 gaps above 1 s, tracker at 1.7 Hz).

Truth	GT path [m]	offset [s]	ATE [m]	median [m]	path ratio
as delivered	36.8	4.38	1.174	0.851	0.65
one sample removed	27.1	-4.14	0.519	0.437	0.85

Analysis. The front end is what separates the runs. The GPU extractor drops a third to a half of the frames and holds 0.5 m; the CPU extractor drops two thirds and loses a factor of two, one of its runs with a 2 m vertical excursion the tool flags as implausible and the other with three resets and a half-second of lost tracking. Loop closing changes nothing that the front end does not change more. The remaining resets are an initialisation effect, two or three `local_mapping_bad_imu` map resets inside the first 13 s and none after (zero in the GPU run without loop closing); after them the error is a slow wave between 0.05 and 1 m, not an accumulating drift (figure 3.8), and it does not rise while turning in any run (table 3.6). The estimates are a uniform 12–15 % smaller than the tracker's path, which the unmeasured lever arm, a stereo-baseline error or a tracker scale error could each explain. One tracker sample decides the headline: scored against the export as delivered, the GPU run reads 1.17 m with a path ratio of 0.65, because that sample adds 10 m of false path and dominates the time alignment (table 3.5).

The setting was reached by elimination. With the IMU noise densities copied from the VINS file the estimator was unstable on this recording, running away in its velocity state at a different place in each run;

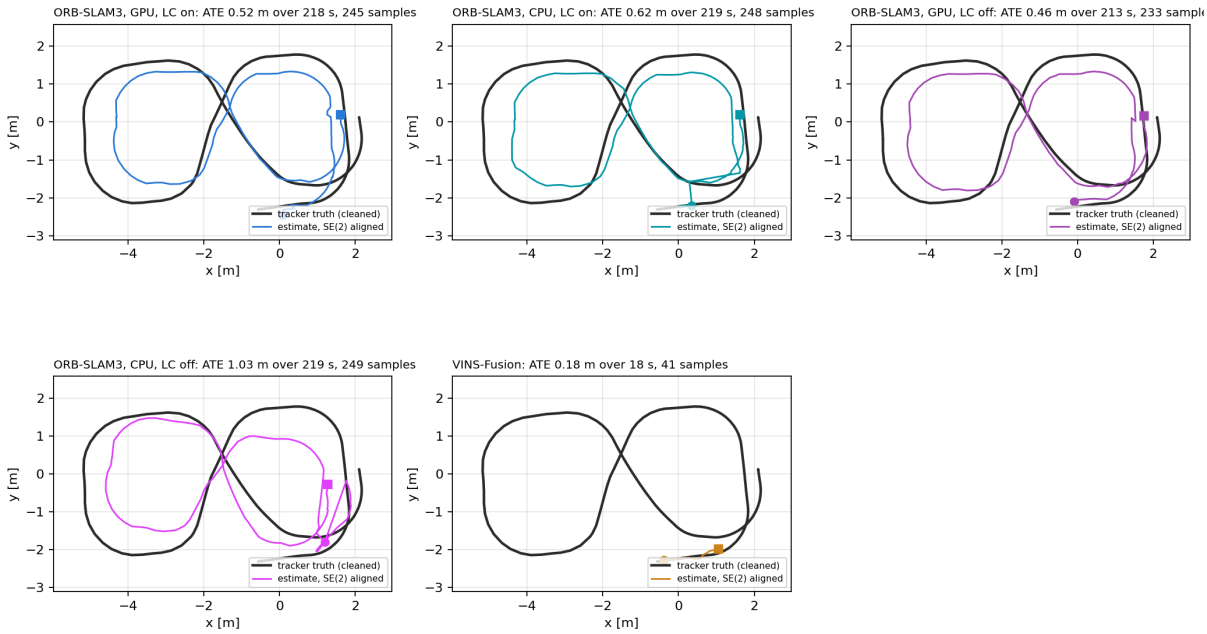


Figure 3.7: Every run against the cleaned tracker truth after rigid SE(2) alignment, one panel per run: ORB-SLAM3 with GPU or CPU extractor and loop closing on or off, and VINS-Fusion's 18 s.

Table 3.6: ORB-SLAM3 on `dataset_realv6_000`, cleaned truth: growth rate of the planar error while the aligned estimate's heading changes faster than $10\,^\circ\,\mathrm{s}^{-1}$ and on straights, with the time in each state. Positive = error growing. One run per configuration, so no test is possible.

Run	turning [mm/s]	straight [mm/s]	time turning [s]	time straight [s]
ORB-SLAM3, GPU, LC on	-11.2	-0.3	23	171
ORB-SLAM3, CPU, LC on	-29.8	-1.4	23	162
ORB-SLAM3, GPU, LC off	4.9	-2.1	11	185
ORB-SLAM3, CPU, LC off	-49.9	-0.9	20	182

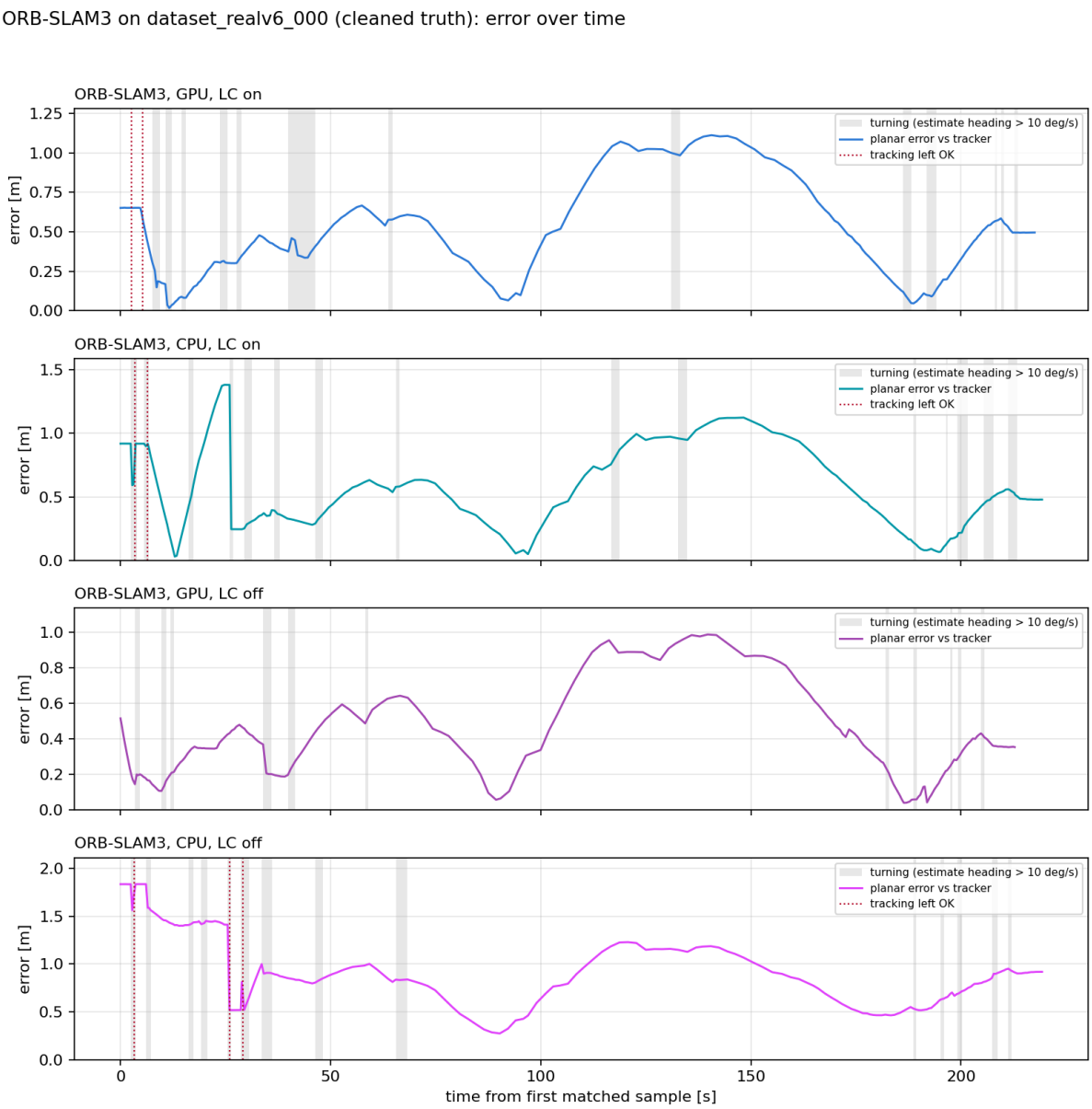


Figure 3.8: ORB-SLAM3 planar error over the run, one panel per configuration; grey bands are turns, dotted red the initialisation resets.

six out-of-bounds reads in the library were fixed without removing it, every other candidate was varied without removing it, and raising the IMU noise densities tenfold removed it in every run since. Those earlier runs were overwritten and carry no number here; the search is in the ORB-SLAM3 engineering record.

VINS-Fusion's failure is a delivery defect. The node received 3 187 of 6 601 frames but used 6 498 of the 43 494 IMU messages it received, reset 217 times on `imu_gap`, and stopped after 19 s; behind that defect the configuration still fixes a gravity norm of 9.2642 against the 9.878 m s^{-2} this IMU reads at rest.

Table 3.7: VINS-Fusion on `dataset_realv6_000`, 3 November 2026 (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm still 9.2642 against the 9.878 m s^{-2} this IMU reads at rest). The scored span is the 18 s it tracked before its last reset and is not a result.

Quantity	Value
camera frames received (bag: 6 601)	3187
IMU messages received / samples used	43494 / 6498
`imu_gap` resets	217
poses written / span of poses	286 / 19 s
tracker samples matched	41 of 43 in span
ATE over the tracked span	0.18 m (not a result)

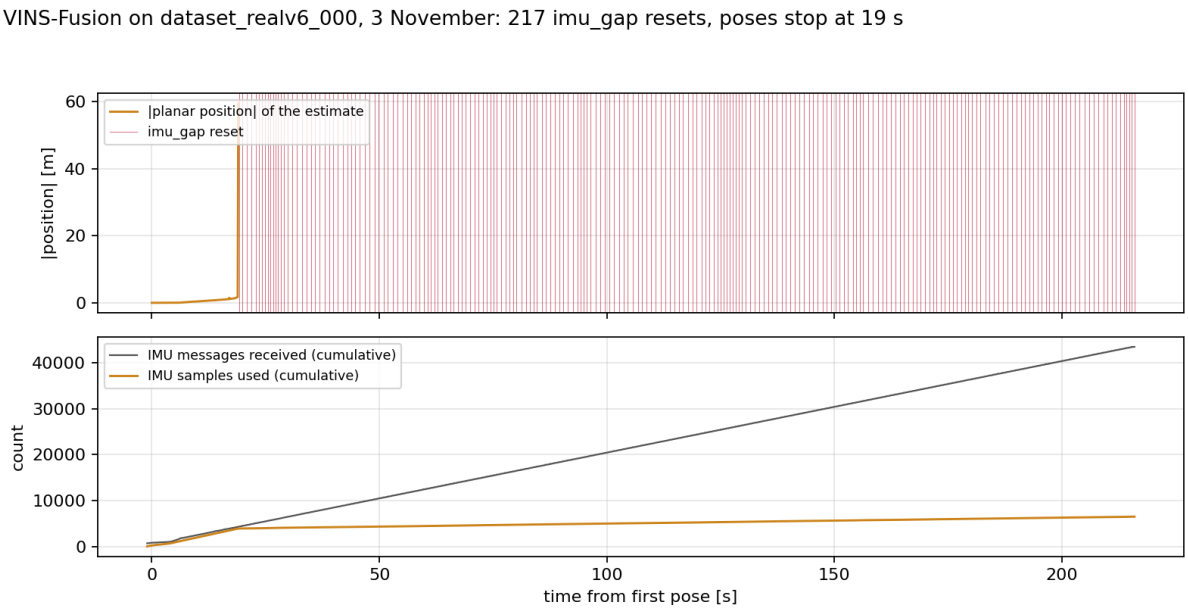


Figure 3.9: VINS-Fusion: distance from origin with every `imu_gap` reset marked, and IMU messages received against samples used.

Second recording set, realv7. Three further recordings of 3 November 2026 (`dataset_realv7_000`, `_001`, `_002`: 21.6, 31.7 and 23.4 m over 97, 134 and 104 s) have tracker exports that pass every validity rule (5–6 Hz, no gap, no step). Each carries the same four ORB-SLAM3 configurations and one VINS-Fusion run. As written in the live log (table 3.8, figure 3.10), a plane fitted to the ORB-SLAM3 estimate lies within 10° of horizontal in three runs of twelve and at 88–90° in five, every `000` run among them, with the plane normal exactly a body axis in every run; scored as written, those nine runs are not scorable.

The per-frame diagnostics give the mechanism. The IMU initialised 3–4 s after the first pose. The upstream rule that resets a map with less than 2 cm between consecutive keyframes in the first ten seconds after the initialisation then discarded it, the robot being still at rest. The new map starts in the camera

frame of the reset instant, and in it the re-initialisation was starved: upstream attempts it only with an empty keyframe queue, which a keyframe every 0.25 s and 250 ms of map-point creation per keyframe never allowed. The IMU re-initialised after 37–92 s in seven runs, never in two, and early (6.7 s) in the one run that was level. Until then the estimate is stereo-only in a map whose vertical is the camera's y axis, and the late re-initialisation rotates the map under the already-written poses (the 1.9–6.8 m steps).

The scorer repairs both without the truth: the pre-initialisation poses are rotated by the rotation measured at the initialisation frame, and each map's fitted plane is levelled. With the bags on the dataset drive the time offsets come from the wheel speed on all three recordings (table 3.9, figure 3.11). `000` scores 0.25–0.27 m in all four configurations over 89 s (RPE at 1 s 0.05 m), with the IMU initialised for at most a quarter of the frames. `001` scores 0.44–0.82 m over 126 s, one run failing the scorer's one-metre vertical-range rule by two centimetres. `002` scores 0.47 m for the CPU front end with loop closing; two runs lose tracking inside the route and one drifts (vertical range 4.3 m, failed). The init-time resets leave every other run *partial* under the scorer's rules.

The fork's patch 15 adds two switches that attempt the initialisation with a queued backlog and skip the low-motion reset for stereo; they are on in the real-rig configuration and not yet exercised on a recording. VINS-Fusion diverges on all three with the IMU stream intact (zero resets) and 40–45 % of the frames received (table 3.10); its gravity norm has since been set to the measured 9.886 m s^{-2}.

Table 3.8: ORB-SLAM3 on the three `dataset_realv7` recordings, 3 November 2026 afternoon, same setting as the realv6 runs: the plane of each estimate as written in the live log, before any repair. "Tilt" is the angle between the plane fitted to the estimate's 3-D positions and the horizontal; "off-plane" the rms distance of the positions from that plane. A floor robot's estimate should be horizontal: a tilt near 90° means the poses were written while the map was not gravity-aligned (IMU not initialised; the vertical is then the camera's y axis). The scores are in table 3.9, after the repair.

Recording	Run	tilt [°]	off-plane [m]	z range [m]	resets
`realv7_000`	GPU, LC on	89.1	0.35	6.87	3 s
`realv7_000`	GPU, LC off	89.5	0.02	6.98	3 s
`realv7_000`	CPU, LC on	88.8	0.41	6.88	5 s
`realv7_000`	CPU, LC off	87.9	0.58	6.79	4 s
`realv7_001`	GPU, LC on	88.8	0.05	9.64	4 s
`realv7_001`	GPU, LC off	6.1	1.12	5.77	3 s, 5 s
`realv7_001`	CPU, LC on	1.5	0.07	0.38	4 s
`realv7_001`	CPU, LC off	13.0	1.05	5.23	5 s
`realv7_002`	GPU, LC on	37.7	0.87	4.08	3 s, 49 s, 59 s
`realv7_002`	GPU, LC off	59.7	0.83	4.80	49 s
`realv7_002`	CPU, LC on	5.2	0.73	2.94	none
`realv7_002`	CPU, LC off	21.7	0.84	7.03	none

[Sparse reference] The tracker ran at 1.7 Hz rather than its nominal 7 Hz on `realv6`, and the validity rules were written for the faster export. Either the export has to be re-captured at full rate or a rule for sparse references has to be written before any value from this recording is quoted outside a diagnostic.

Summary. The real robot has an unfiltered ORB-SLAM3 reference at a provisional 0.46–0.52 m on `realv6` (GPU front end) and 0.25–0.27 m on `realv7_000` with valid truth after the scorer's frame repair, a measured cost of the CPU front end at real time, the cause of the IMU staying uninitialised for most of nine `realv7` runs (an upstream map reset at the initialisation and a starved re-initialisation) patched in the fork but not yet exercised, and two diagnosed VINS-Fusion defects, one of them (the gravity norm) now corrected in the configuration. The experiment's own question, whether filtering helps, is still untested on real imagery.

Next action. Run one `realv7` recording with the fork's two initialisation switches on and confirm from the per-frame log that the IMU stays initialised from the first seconds; re-run VINS-Fusion with

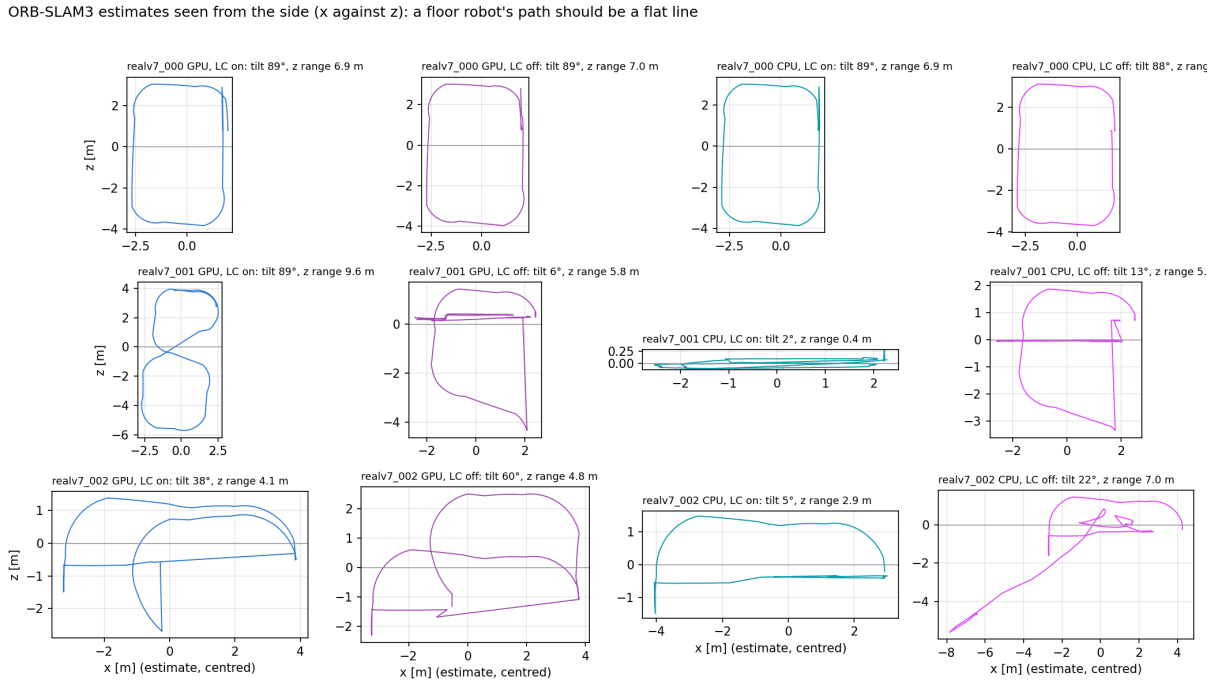


Figure 3.10: The twelve `realv7` ORB-SLAM3 estimates as written in the live log, seen from the side (x against z, centred); a floor robot's path should be a flat line. The repaired estimates are in figure 3.11.

Table 3.9: ORB-SLAM3 on the three `dataset_realv7` recordings (3 November 2026, same setting as realv6), scored on 4 November with the bags on the dataset drive (wheel-speed time offsets) and the two frame repairs of `script/eval_real_gt.py`: the poses written before each IMU initialisation rotated into the frame used after it (`-join-imu-init`), and each map's fitted plane levelled (`-level-plane`). "IMU" is the share of tracked frames with the IMU initialised; "resets" the active-map resets after the first pose, in seconds (a reset restarts the position at the origin and is not stitched; those inside the route are tracking losses); "tilt" the plane tilt of the raw log, zero after the repair in every run. ATE is planar after SE(2) alignment, RPE over 1 s. Status from the scorer's rules ("partial" = a reset or loss the run recovered from). Lever arm unmeasured (zero); VINS-Fusion diverged on all three (table 3.10) and is omitted.

Recording	Run	IMU [%]	resets [s]	tilt [°]	ATE [m]	median [m]	$\text{RPE}_{1\,\text{s}}$ [m]	path ratio	status
`realv7_000`	GPU, LC on	6	3	89	0.25	0.24	0.05	1.05	partial
`realv7_000`	GPU, LC off	0	3	89	0.26	0.21	0.05	1.09	partial
`realv7_000`	CPU, LC on	11	5	89	0.27	0.21	0.05	1.06	partial
`realv7_000`	CPU, LC off	23	4	88	0.26	0.23	0.05	1.08	partial
`realv7_001`	GPU, LC on	3	4	89	0.48	0.47	0.07	1.07	partial
`realv7_001`	GPU, LC off	67	3, 5	6	0.82	0.63	0.15	1.15	partial
`realv7_001`	CPU, LC on	94	4	2	0.44	0.40	0.07	1.07	partial
`realv7_001`	CPU, LC off	65	5	13	0.56	0.50	0.10	1.11	failed
`realv7_002`	GPU, LC on	22	3, 49, 59	38	2.11	1.28	0.26	1.18	partial
`realv7_002`	GPU, LC off	22	49	60	2.08	1.83	0.23	1.29	partial
`realv7_002`	CPU, LC on	61	none	5	0.47	0.41	0.12	1.13	partial
`realv7_002`	CPU, LC off	62	none	22	1.67	0.98	0.70	1.68	failed

Table 3.10: VINS-Fusion on the three `dataset_realv7` recordings (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm 9.2642). Frames received against the bag's count (30 Hz over the recording), IMU use, resets, and where the estimate ended.

Recording	frames received	IMU received / used	resets	z range [m]	$\lvert p \rvert$ at end [m]
`realv7_000`	1155 of about 2663	17906 / 19053	0	12834403	37554
`realv7_001`	1541 of about 3686	24809 / 26340	0	3701	15974
`realv7_002`	1271 of about 2910	19368 / 20627	0	25937	5252

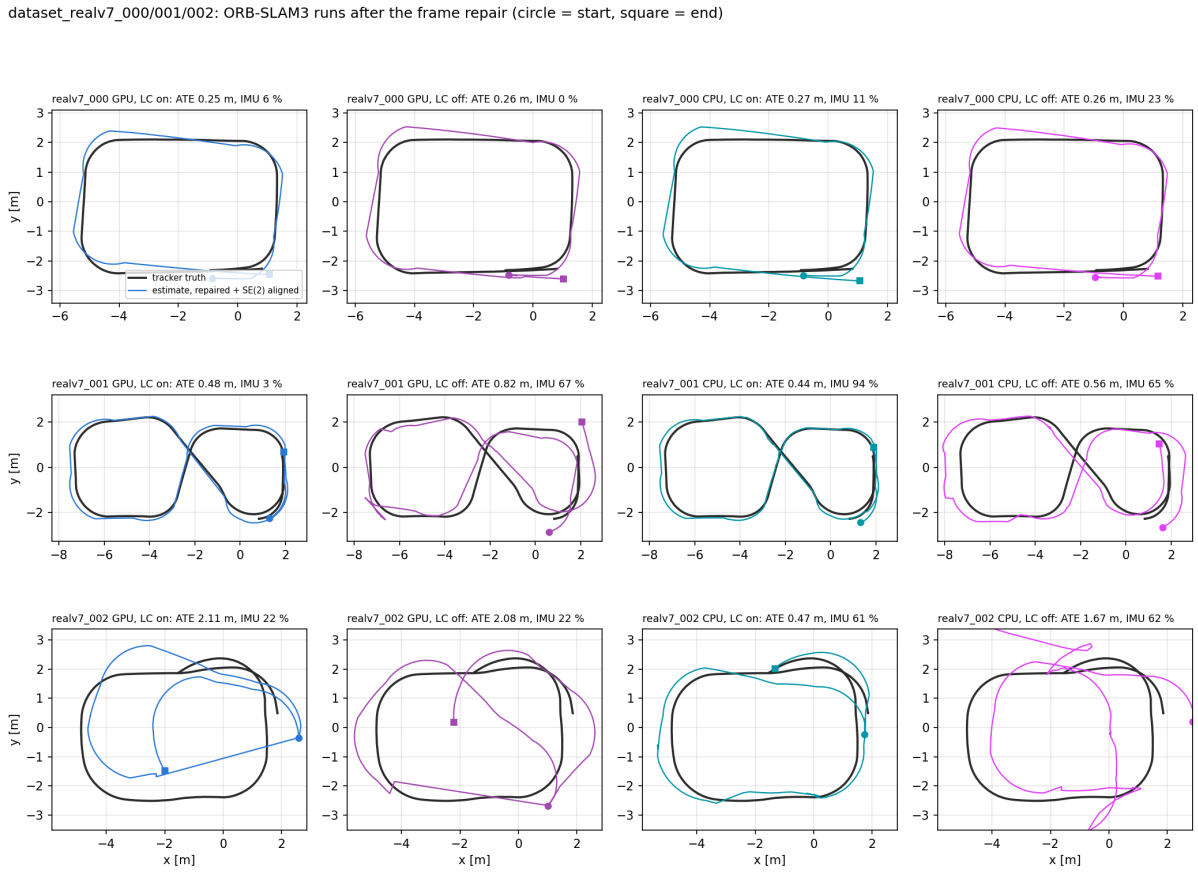


Figure 3.11: The twelve `realv7` ORB-SLAM3 runs after the scorer's frame repair, against the tracker truth after rigid SE(2) alignment; each panel's title gives the planar ATE and the share of tracked frames with the IMU initialised.

the corrected gravity norm; repeat each ORB-SLAM3 configuration under a new run name; measure the marker-to-IMU lever arm; then run both pipelines filtered and unfiltered on the same recording, with the detector live, and score them with the admissibility gates of the simulation slots.